PROJECT PROPOSAL

Small Business Sales and Profit Tracking System Using Oracle PL/SQL

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Database: Oracle Database 26ai Free

GitHubRepository:https://github.com/Bonnette26/Niyigena_20252SEN266_SmallBusinessSalesProfit

1. Problem Statement

Many small businesses face challenges in managing daily sales, monitoring product stock, recording operating expenses, and calculating profits accurately. When these activities are managed manually, businesses may experience calculation errors, missing records, poor inventory control, and difficulties understanding their financial performance.

There is a need for a reliable database system that organizes business information, automates important calculations, and provides useful reports to support decision-making.

2. Proposed Solution

This project will develop a **Small Business Sales and Profit Tracking System** using Oracle Database and PL/SQL. It will organize products and suppliers, record sales and sale details, monitor stock, and record operating expenses. PL/SQL program units will apply business rules, validate transactions, handle errors, automate selected updates, and support reports. Revenue and estimated profit will be calculated from recorded sales, documented cost assumptions, and expense data.

3. Data Source

The project uses the **UCI Online Retail dataset** (Chen, 2015; DOI: 10.24432/C5BW33), containing 541,909 transaction records from a UK-based online retailer. Its fields include invoice number, stock code, product description, quantity, invoice date, unit price, customer identifier, and country. The dataset is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). The original records are retained in a staging table for traceability; cancellations, returns/negative quantities, and missing values are identified and documented rather than silently discarded. The source does not provide product purchase costs or operating expenses. Any additional cost ratios or expense examples used for demonstrations will be clearly labelled as assumptions or illustrative project data, not as original dataset facts. Customer identifiers will not be used to identify individuals in reports.

4. Planned Tables

The database design contains eight main tables:

SUPPLIERS: Stores supplier information related to products.

CUSTOMERS: customer records and source identifiers.

PRODUCTS: Stores product names, descriptions, prices, costs, and available stock quantities.

USERS: Stores system user information and roles for managing access.

SALES: Stores general sales transaction information, including sale dates and transaction totals**.**

SALE_DETAILS: Stores individual products, quantities, and prices belonging to each sale. It connects SALES and PRODUCTS.

EXPENSES: Records business operating expenses, amounts, categories, and dates.

AUDIT_LOG: Records selected database changes for accountability and tracking.

These tables will be connected through primary and foreign keys to maintain data consistency and relationships.

5. Planned PL/SQL Features

Procedures will register products, record sales, update stock, and record expenses.

Functions will calculate revenue, costs, expenses, and estimated profit.

Cursors will retrieve records for reports.

Packages will group related reusable program units.

Triggers will automate selected stock updates and audit logging.

Exception handling will manage invalid quantities, insufficient stock, missing records, and other errors.

Collections and bulk processing will support efficient operations on groups of records.

6. Innovation

The planned innovation is an interactive **Microsoft Power BI analytics dashboard** connected to Oracle Database 26ai Free through the Oracle connector/client or ODBC driver. It will visualize sales, expenses, estimated profit, monthly trends, best-selling products, and inventory levels. Power BI will present database information, while core business rules and calculations remain implemented in PL/SQL. The connection method, setup, and evidence will be documented in a later project phase.

7. Expected Results

The expected result is a working Oracle PL/SQL database system that organizes products, suppliers, sales, sale details, expenses, users, and audit records. The system should support accurate transaction recording, stock monitoring, estimated profit calculations, and useful business reports.

8. References

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Mintah, C., Gabir, M., Aloo, F., & Ofori, E. K. (2022). Do business records management affect business growth? *PLOS ONE, 17*(3), e0264135. https://doi.org/10.1371/journal.pone.0264135

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